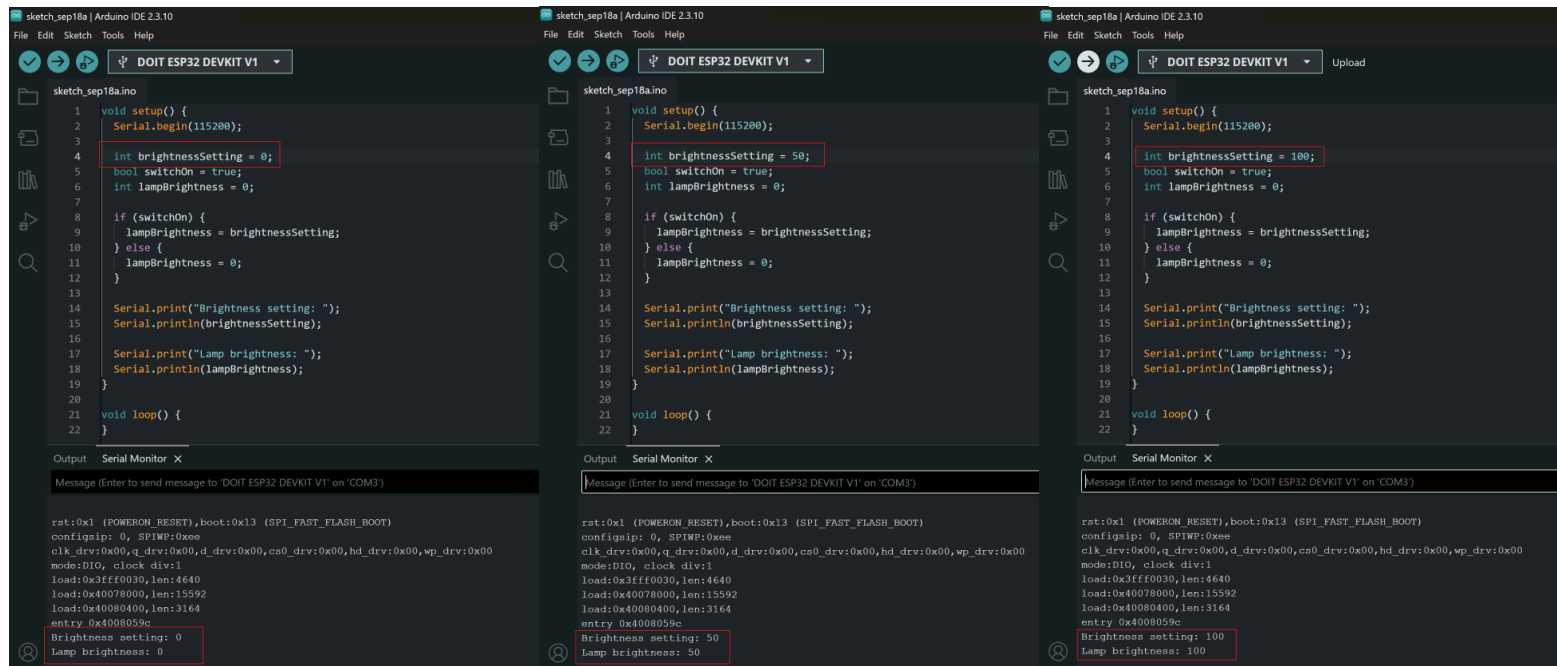
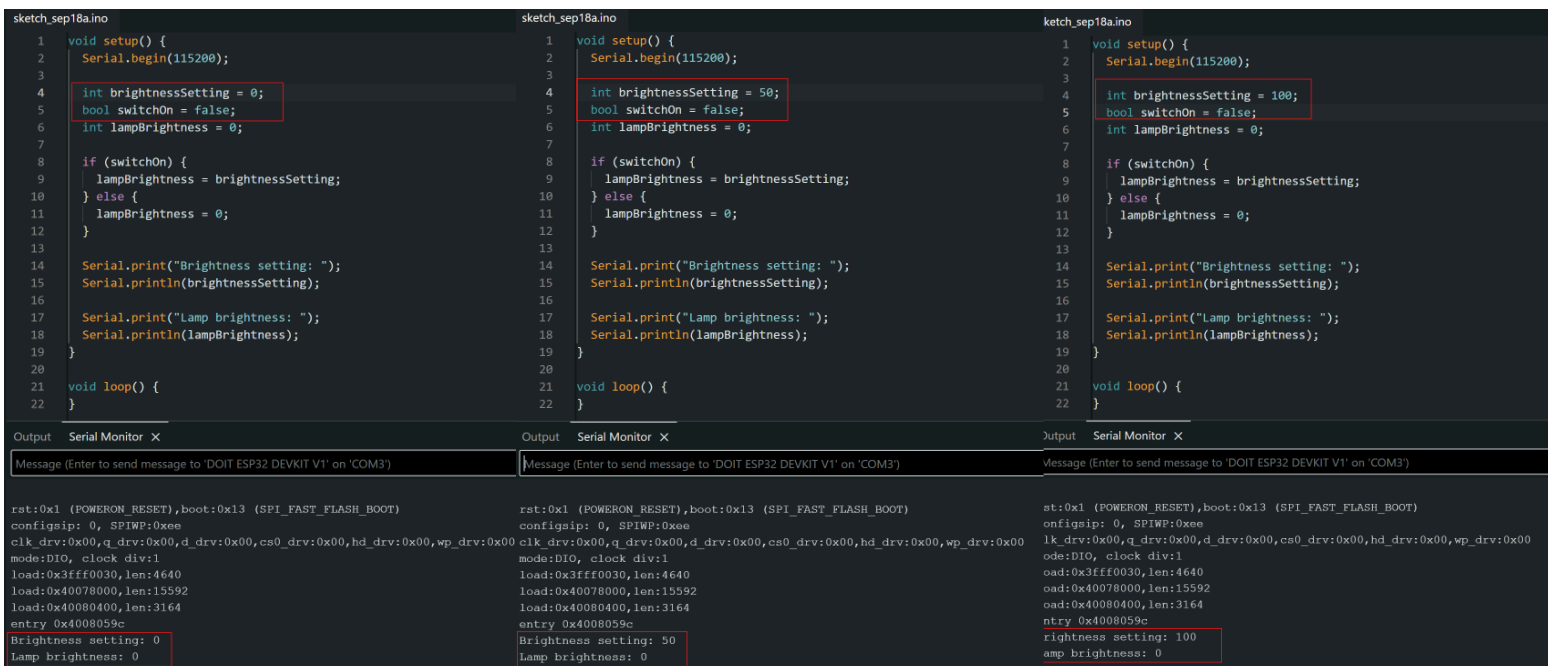


1. Set `brightnessSetting` to 0, 50, and 100. Run the program for each value with `switchOn` set to `true`.



2. Repeat the tests with `switchOn` set to `false`.



Brightness setting	Switch state	Expected lamp brightness	Observe output
0	On	0	0
50	On	50	50
100	On	100	100
0	Off	0	0
50	Off	0	0
100	Off	0	0

4-5.

```

sketch_sep18a.ino
1  int getLampBrightness(bool switchOn, int brightnessSetting) {
2      if (switchOn) {
3          return brightnessSetting;
4      } else {
5          return 0;
6      }
7  }
8
9  void setup() {
10     Serial.begin(115200);
11
12     int brightnessSetting = 50;
13     bool switchOn = true;
14
15     int lampBrightness = getLampBrightness(switchOn, brightnessSetting);
16
17     Serial.print("Brightness setting: ");
18     Serial.println(brightnessSetting);
19
20     Serial.print("Lamp brightness: ");
21     Serial.println(lampBrightness);
22 }

```

Output Serial Monitor X

Message (Enter to send message to 'DOIT ESP32 DEVKIT V1' on 'COM3')

```

rst:0x1 (POWERON_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
configsip: 0, SPIWP:0xee
clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
mode:DIO, clock div:1
load:0x3fff0030,len:4640
load:0x40078000,len:15592
load:0x40080400,len:3164
entry 0x4008059c
Brightness setting: 50
Lamp brightness: 50

```

The `getLampBrightness()` function has two parameters: `switchOn` and `brightnessSetting`. The `switchOn` parameter represents whether the switch is on or off, while `brightnessSetting` represents the selected brightness value. The function returns the brightness setting when the switch is on and returns 0 when the switch is off.

Coding Check

1. In `int brightnessSetting = 75;`, identify the data type, variable name, and initial value. How would you change the value to 50?

The data type is *int*, the variable name is *brightnessSetting*, and the initial value is 75. To change it to 50 we need to set the *int brightnessSetting = 50*.

2. Why is `bool` suitable for `switchOn`, while `int` is suitable for `brightnessSetting`?

The *bool* is suitable for *switchOn* because it only needs two values: The *true* for ON and *false* for OFF. Also, the *int* is suitable for *brightnessSetting* because it stores whole numbers like 0, 50, and 100.

3. Predict the output when `brightnessSetting` is 80 and `switchOn` is `false`.

Brightness setting: 80

Lamp brightness: 0

This is because when the switch is on, the lamp brightness matches the setting and when the switch is off, the lamp brightness is zero.

4. Are `switchOn` and `switchon` the same variable? Explain.

No, because programming language is strict when it comes to case sensitivity. In result, the `switchOn` and `switchon` are considered different variable names because 'O' and 'o' have different capitalization.

5. What values are passed to `getLampBrightness()`, and what does it return?

The values of `switchOn` and `brightnessSetting` are passed to the function:

`getLampBrightness(switchOn, brightnessSetting);`

Because the function returns the brightness setting when `switchOn` is true. If `switchOn` is false, it will return 0.

6. If `lampBrightness` is declared inside `setup()`, can `loop()` access it directly? Explain.

No. The `loop()` cannot access it directly because `lampBrightness` is declared inside `setup()`, making it a local variable. It can only be used inside `setup()`.